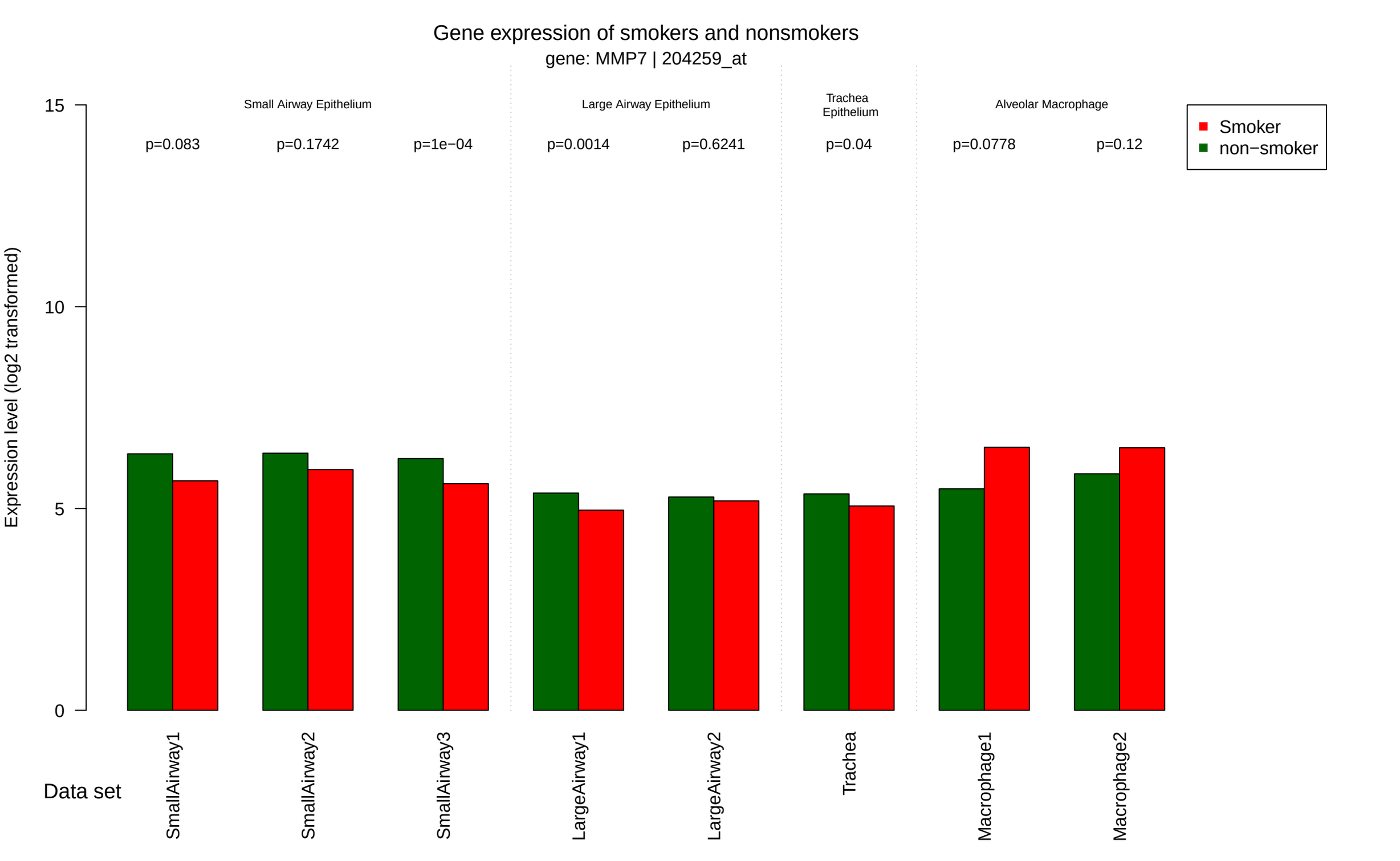
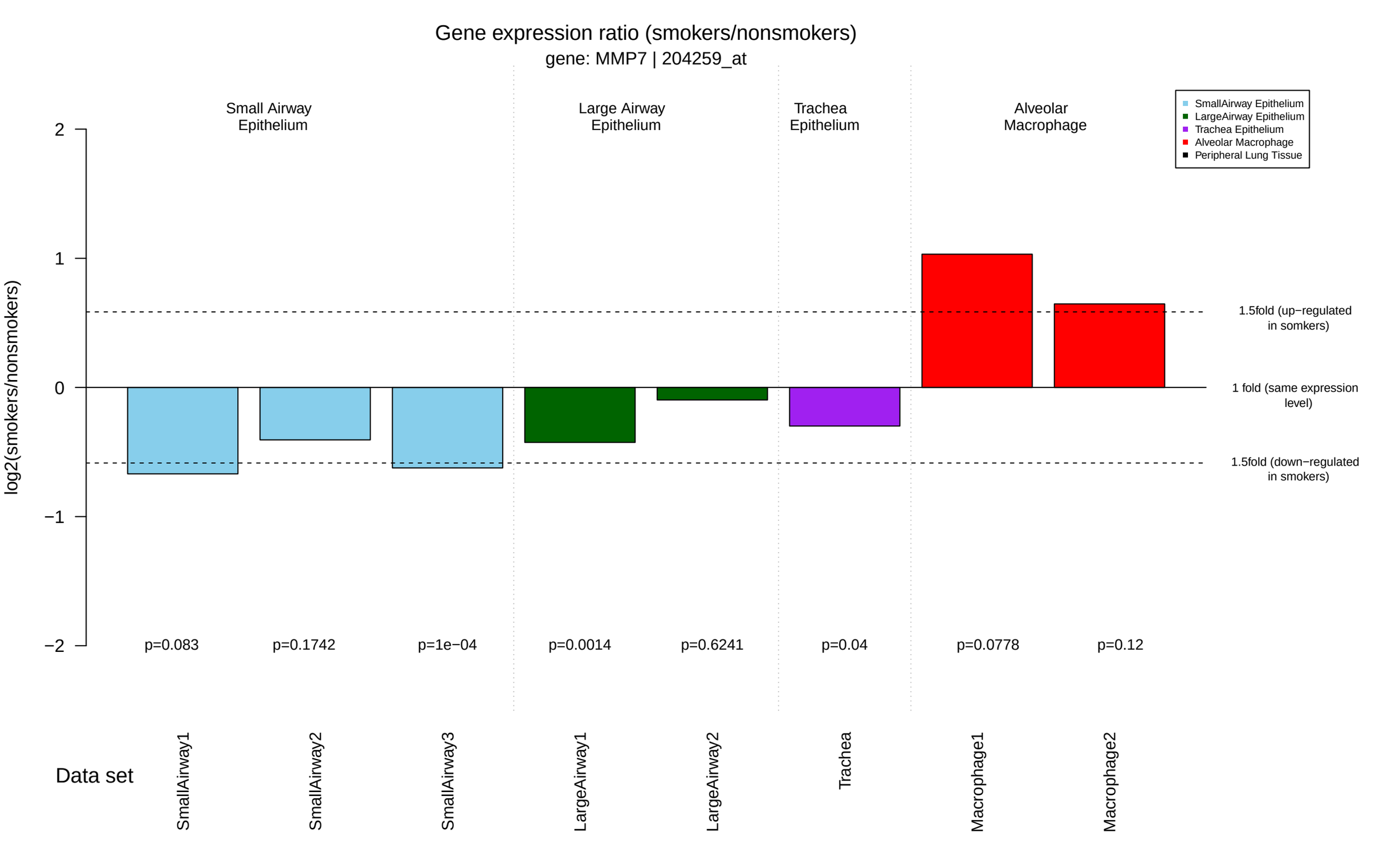
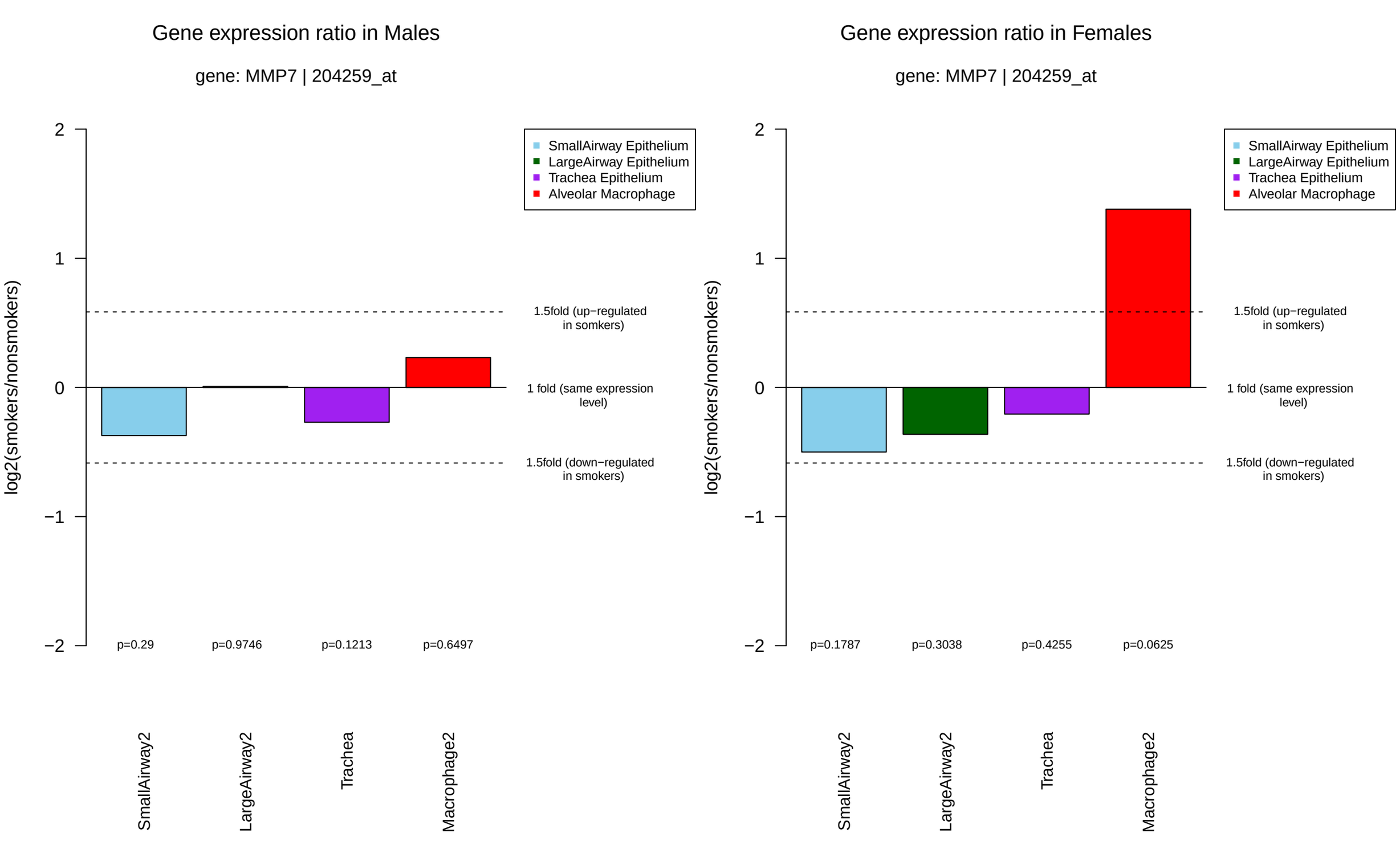
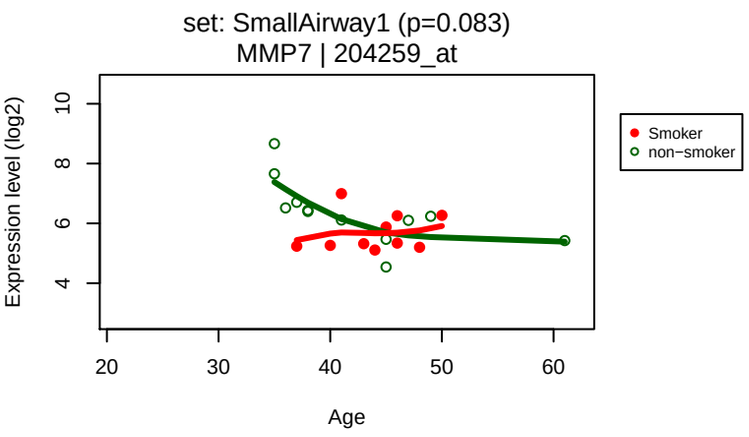
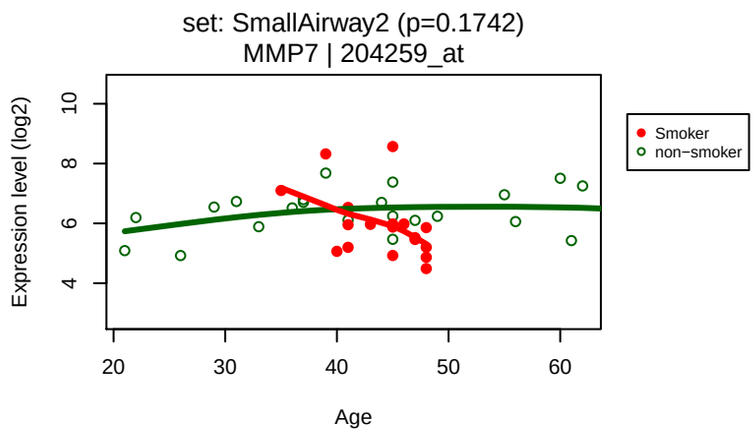
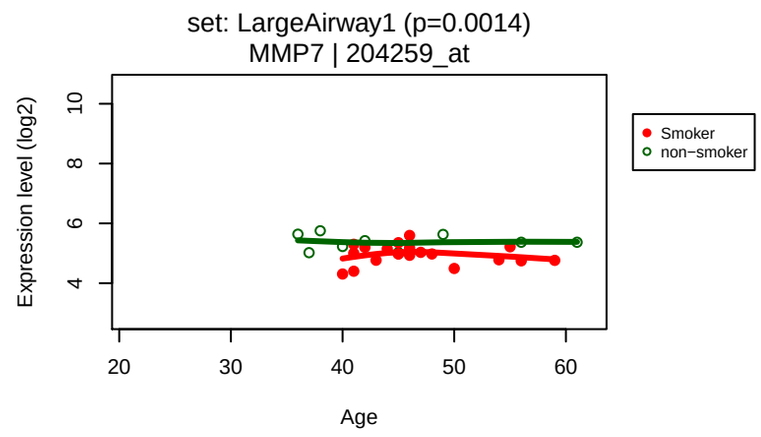

Gene expression of smokers and nonsmokers
gene: MMP7 | 204259_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.083
p=0.1742
p=1e−04
p=0.0014
p=0.6241
p=0.04
p=0.0778
p=0.12
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

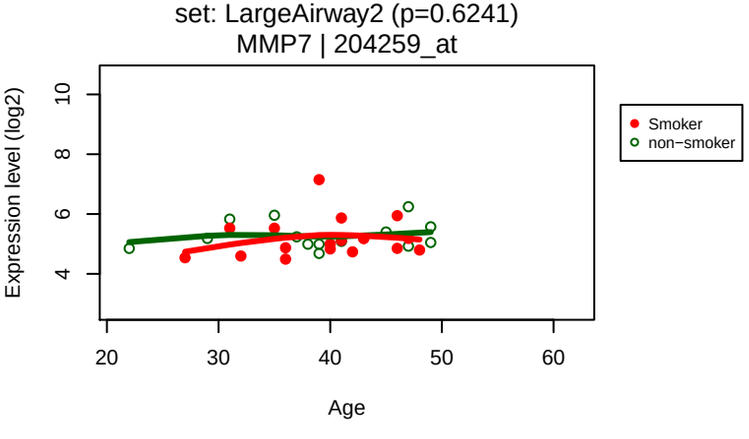

Gene expression ratio (smokers/nonsmokers)
gene: MMP7 | 204259_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.083
p=0.1742
p=1e−04
p=0.0014
p=0.6241
p=0.04
p=0.0778
p=0.12
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

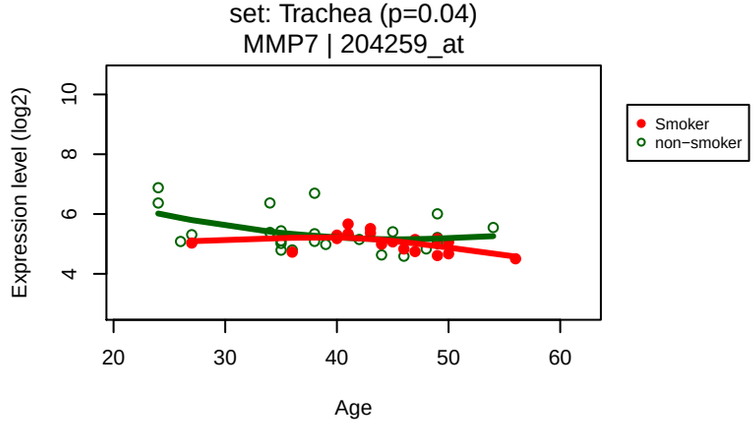

Gene expression ratio in Males
gene: MMP7 | 204259_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.29
p=0.9746
p=0.1213
p=0.6497
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MMP7 | 204259_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1787
p=0.3038
p=0.4255
p=0.0625
SmallAirway2
LargeAirway2
Trachea
Macrophage2

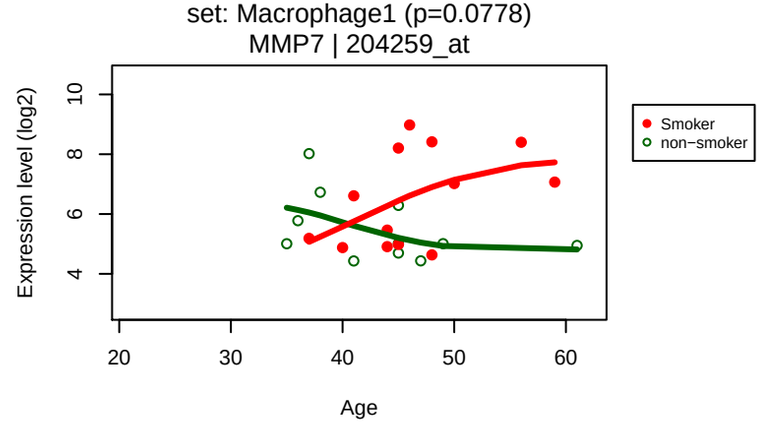

set: SmallAirway1 (p=0.083)
MMP7 | 204259_at
Expression level (log2)
Age
Smoker
non−smoker

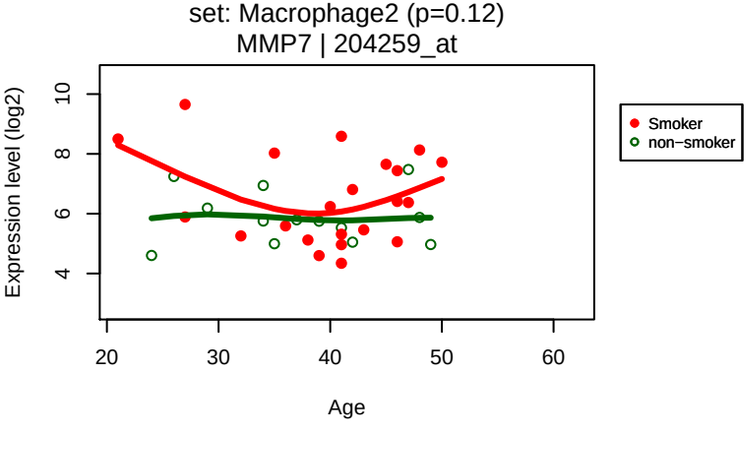

set: SmallAirway2 (p=0.1742)
MMP7 | 204259_at
Expression level (log2)
Age
Smoker
non−smoker

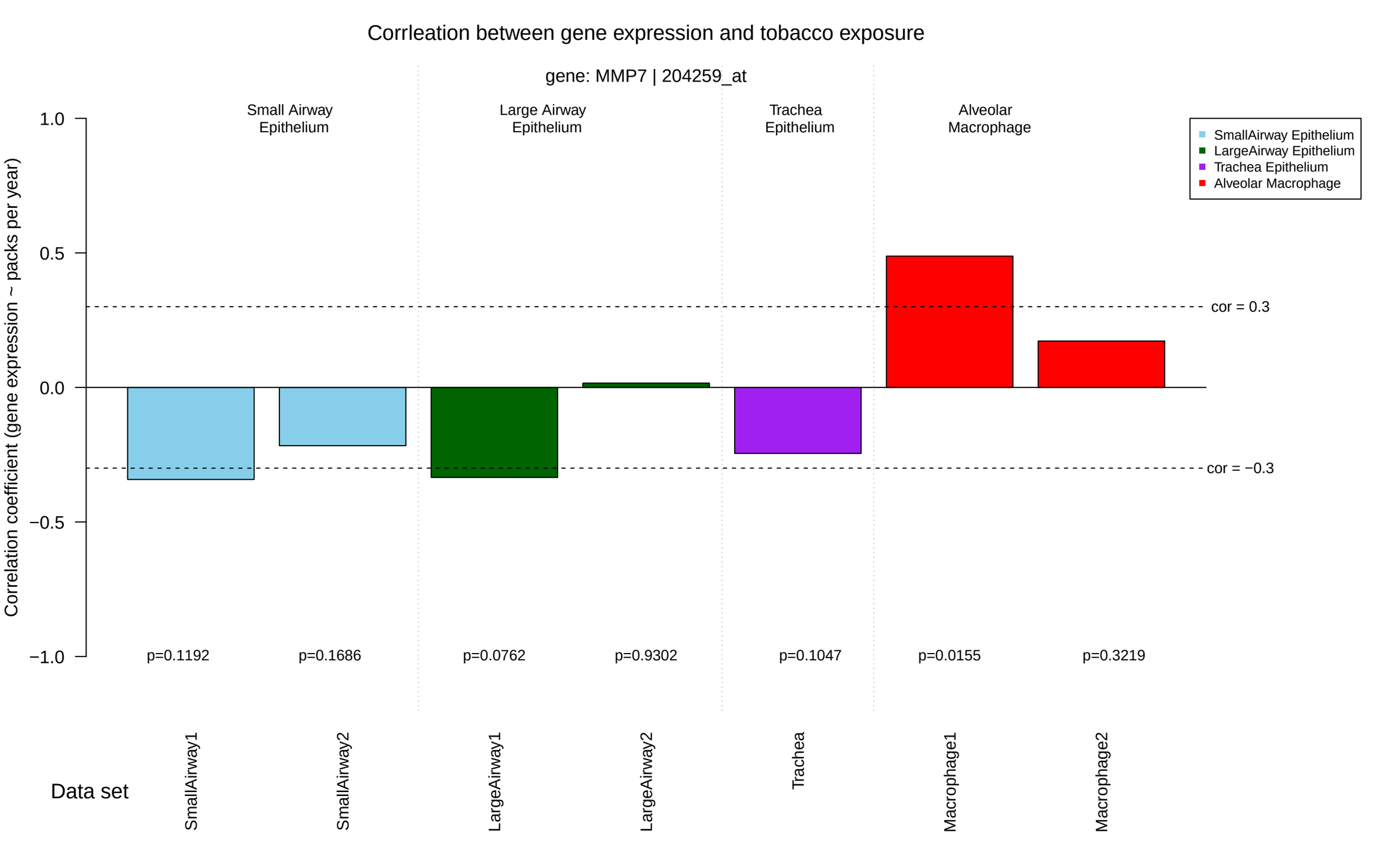

set: LargeAirway1 (p=0.0014)
MMP7 | 204259_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6241)
MMP7 | 204259_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.04)
MMP7 | 204259_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0778)
MMP7 | 204259_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.12)
MMP7 | 204259_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MMP7 | 204259_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1192
p=0.1686
p=0.0762
p=0.9302
p=0.1047
p=0.0155
p=0.3219
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2